1. In GB3 if every 4th packet is lost & we need to send 10 packets so how many retransmissions are required?
2. In SR $W_s = 4$ and we are sending 10 packets where every 4th packet is lost find number of retransmissions?
3. Calculate the sequence number for 68th frame? m=5bits → seq no=0 to 31
4. If Q is the size of sender window in Go Back N ARQ, calculate the number of sequence bits.
5. If the bandwidth of the channel is 10Mbps, calculate the 1 bit delay in time.
6. If $v=2*10^8$m /sec then calculate the 1 bit delay in terms of meters of cable?
7. Compute approximate optimal window size when packet size is 53 bytes, RTT is 60 msec and bottleneck bandwidth is 155 Mbps.
8. In GBN sender Window size = 10 and $T_p = 49.5$ms & $T_t = 1$ms. What is the Efficiency of the protocol and Throughput, given Bandwidth = 1000 bps?
9. A sliding window protocol is designed for a 1 Mbps point to point link to the moon which has a one way latency (delay) of 1.25 sec. Assuming, that each frame carries 1 KB of data, what is the minimum number of bits needed for the sequence number?
10. Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 μs. Acknowledgement packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 μs. What is the maximum achievable throughput in this communication?
11. Station A uses 32 byte packets to transmit messages to station B using a sliding window protocol. The round trip delay between A and B is 80 msec and the bottleneck bandwidth on the path between A and B is 128 Kbps. What is the optimal window size that A should use?
12. A 3000 km long trunk operates at 1.536 Mbps and is used to transmit 64 byte frames and uses sliding window protocol. If the propagation speed is 6 μsec / km, how many bits should the sequence number field be?
13. Two peer process A and b use stop and wait ARQ to send packets over a single channel with capacity C. All packets have the same length of 100 bits. The round trip time (RTT) which is the time until A receives an acknowledgment for a sent packet is equal to 2 sec. Assume that no packet or acknowledgement are dropped and that all packets and acknowledgement arrives error free. Furthermore assume that the capacity is equal to 100000 bps. Find the average transmission rate (in bps) with process A sends data to process B. What is the link utilization?
14. The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the minimum

number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

(a) log2 [(2LtR + 2K)/K]
(b) log2 [(2LtR)/K]
(c) log2 [(2LtR + K)/K]
(d) log2 [(2LtR + K)/2K]